Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/29/2006
PI Name:	Hollberg, Leo W Ph.D.		
Project Title:	Advanced Optical Frequency Standard for Space		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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PI Email:	lhollberg@earthlink.net	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	303-582-4568
Organization Name:	Stanford University		
PI Address 1:	Physics Department		
PI Address 2:	452 Lomita Mall, HEPL		
PI Web Page:			
City:	Stanford	**State:**	CA
Zip Code:	94305-4008	**Congressional District:**	15
Comments:	NOTE: Previously at National Institute of Standards & Technology (Ed., Fall 2011)		
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Maleki, Lute	**Contact Phone:**	818-354-3688
Contact Email:	lute.maleki@jpl.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	C73540A		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		
	To develop high-accuracy high-stability optical frequency standards for space based on laser-cooled and trapped neutral atoms and ions. Microwave frequency standards have achieved higher stability and accuracy in the last decade due in large part to laser cooling and trapping [1,2,3]. Better stability and accuracy will be difficult since microwave clocks are now near their fundamental performance limit. However, if the frequency of the standard could be made higher, then its stability (and possibly its accuracy) could be proportionally higher [4]. Thus, optical oscillators that are referenced to narrow resonances in laser-cooled atoms will be the time and frequency standards of the future. Our research focuses on eliminating or reducing the most serious impediments to the performance of optical frequency standards. These include: 1.Acceleration-induced noise in optical reference cavities 2.Gravitation limitation to the interaction time for neutral-atom clocks 3.An optical frequency synthesis/measurement system that preserves the phase fidelity of the optical references		

Task Description:	We are evaluating the performance of two optical frequency standards, one based on a trapped and laser-cooled single Hg+ ion and the other on laser-cooled calcium atoms. Since the fractional frequency instability of atomic clocks is inversely proportional to the frequency of the clock transition and to the square root of the Ramsey interrogation period, greatly improved performance compared to microwave frequency standards is possible by utilizing a narrow optical transition in laser-cooled atoms. Projected fractional frequency instabilities and inaccuracies for laser-cooled, stored ion optical frequency standards that are earth based are about 1x10-18, more than 2 orders of magnitude better than that projected for atomic microwave frequency standards even in a micro-gravity environment. However, optical frequency standards based on laser-cooled neutral atoms in a micro-gravity environment could potentially produce even higher stabilities. In order to realize their projected stabilities, both types of optical frequency standards require a spectrally pure local oscillator that has a short term (<10 s) instability less than 5x10-16. Most often, this is achieved by locking the frequency of the laser to a high-finesse and well-isolated optical cavity. Frequency fluctuations of the optical oscillator are then dominantly caused by seismic perturbations that are coupled through the suspension system which must support the cavity against the pull of gravity. Since the fundamental resonance of the isolation support system is proportional to the sqrt(g, seismic isolation could be more than 3 orders of magnitude better in a micro-g environment. Finally, the full benefits of a highly stable and accurate optical frequency standard are available only by counting the zero crossings of the frequency of the optical standard to generate time. We plan to thoroughly investigate broadband optical synthesis schemes using pulsed lasers to connect visible frequencies to countable microwave frequencies in a phase coherent manner. Recent progress in our laboratory and others shows that mode-locked fs-lasers coupled to photonic fibers can be used to span large frequency intervals in the visible spectrum and to measure optical frequencies with high precision and accuracy. The many orders of magnitude improvement expected in the performance of frequency standards based on optical transitions will open many new opportunities in experimental fundamental physics, including: improved tests of relativity, searches for time variation of fundamental constants, precision time and frequency metrology. We can also anticipate applications in advanced communication systems and navigation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have made good progress in FY06 on our NASA funded project for optical frequency standards for space. We have continued to make steady strides in the operation and evaluation of an all-optical atomic clock based on the narrow 2S1/2 - 2D5/2, electric-quadrupole allowed transition in a single, laser-cooled 199Hg+ ion. In the past year we have compared its frequency to that of the Al+/Be+ “quantum logic” optical clock and achieved a fractional frequency instability of about 4 x 10(-17) at 30 000 s, still limited (but very soon to be bypassed) by the present performance of the quantum logic clock. This is an order of magnitude better stability than we reported a year ago, and an order of magnitude better than any other frequency standard. We expect to demonstrate fractional frequency instabilities and inaccuracies < 10(-17) in the near future. We have also continued to make periodic absolute frequency measurements of the mercury optical clock in direct comparison to the Cs fountain standard NIST-F1. The combined statistical and systematical uncertainties in our latest measurement (March, 2006) were < 1 × 10(-15), however the total systematic uncertainty for the mercury standard is now conservatively estimated to be < 3 × 10(-17), more than a factor 10 smaller than the systematic uncertainty of NIST-F1. Presently, the largest systematic uncertainty is due to the AC Zeeman shift derived from any asymmetry in the RF currents delivered to the trap electrodes. We expect to measure this shift soon, but have conservatively set its theoretical upper bound to be <2 × 10(-17). The relative frequency of the two standards has been intermittently measured since September, 2001 and has varied by no more than 1 part in 1014 over that time period. The last three measurements, made over the course of the past year, vary by less than 1 part in 1015. We have also continued to make good progress with the two-ion, “quantum logic” optical clock. As a reminder, in this system two ions of different species are held in a linear RF trap. One ion (the “clock”) provides a suitable optical transition for an atomic clock, whereas the other ion (the “control”) provides a strongly allowed transition for laser cooling the two-ion ensemble as well as the rapid state preparation and detection of the clock ion. In our implementation, a laser-cooled beryllium ion is trapped in a linear Paul trap simultaneously with an aluminum ion. During the last year, we have tried to optimize the cooling process and coherent state manipulation of the beryllium ion toward demonstrating a robust operation of the Al+/Be+ optical clock. Our choice of 27Al+ is motivated by its potential as a high-accuracy optical clock: the 1S(0) <--> 3P(0) clock transition has a very high Q-factor f/delta f approx 10(17) (lambda approx 267 nm) and a small black-body radiation shift. We have now fully implemented quantum logic based spectroscopy to operate the 27Al+ frequency standard and stabilized a narrow- linewidth (< 3 Hz) interrogation laser to the mean transition frequency of the mF = ±5/2, delta mF = 0 clock transitions. And, as reported above, we have measured the ratio of its frequency to that of the single mercury ion optical frequency standard, as well as hydrogen masers maintained by NIST. For the optical clock based on laser-cooled neutral Ca atoms, we have continued our development of a simplified, transportable version of the clock, which could find numerous applications where high stability is critical. Our apparatus now contains just two semi-conductor based laser systems, one for laser cooling at 423 nm, and a second whose frequency probes and then is locked to the clock transition at 657 nm. Initial measurements against a second stable optical standard (to be described) showed an upper limit for the fractional frequency instability of 3 x 10(-15) for a 1 s averaging time with a value in the mid 10(-16)’s at 200 s, values competitive with the best existing standards. We have made considerable progress on a second optical clock, based on laser-cooled neutral Yb, which has enormous potential for high accuracy and stability. We have achieved the first spectroscopy of the 1S(0)- 3P(0) clock transition at 578 nm with atoms confined to an optical lattice. After two stages of laser cooling, we load roughly 10,000 atoms into a 1-D lattice, which is tuned to its Stark shift-cancelling wavelength near 759 nm. With light from a highly stabilized dye laser, we have measured spectroscopic linewidths as narrow as 4 Hz in the optical domain. An important simplification to the lattice clock approach was our use of an even isotope rather than an odd isotope (as is used in other lattice experiments), which was made possible by the addition of a small magnetic bias field. The use of the even isotope greatly reduces the magnetic field sensitivity of this clock, and additionally simplifies spectroscopy of the clock transition.
Bibliography Type:	Description: (Last Updated: 09/03/2025)

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